SAL College of Engineering

CE IT Department

Computer Organization & Architecture (**BE04000251**)

Assignments

Assignment -1

1 Explain how complement number system is useful in computer system. Discuss any one complement number system with example**.**

2 Explain Common bus system.

3 Define RTL. Explain how register transfer takes place in basic computer system. What do you mean by register transfer? Explain in detail. Also discuss three-state bus buffer.

4 Draw and explain basic computer instruction formats. Write a detailed note on instruction cycle with neat diagrams.

5 What is an assembler? With clear flowcharts for first and second pass, explain its working.

6 Draw Diagram of Control Unit of Basic Computer.

7 Write an assembly language program to add 10 numbers from memory.

Assignment -2

1 Draw a flowchart for first pass of an assembler and explain the same in brief.

2 Write a brief note on:- Subroutine call and return – RISC.

3 Write a program to evaluate the following arithmetic statement X= [A * (B + C) – D] / (E + F - G)
(i) using a general register computer with three-address instructions,
(ii) using an accumulator type computer with one-address instructions,
(iii) using a stack organized computer with zero-address operation instructions.

4 Explain following instruction 1.LDA 2.ADD 3.AND 4.CLA

5 Explain following instruction 1.BUN 2.BSA 3.CIL 4.SZE.

6 What is instruction cycle? Draw Flowchart for Instruction cycle and explain it.

7 For the following C language code , write assembly language program: int a, b, c;
a = 83; // plus 83
b = -23; //minus 23
c=a + b;

Assignment -3

1 Write a note on micro program sequencer.

2 Describe micro programmed control organization in detail.

3 Enlist major components of CPU.

4 State the major characteristics of RISC processor.

5 Explain pipelining technique. Draw the general structure of four segment pipeline.

6 Elaborate Flynn's classification for computers.

7 Draw flow chart for multiplication of two floating point numbers.

8 Explain Booth multiplication algorithm for multiplying binary integers in signed 2's complement representation.

Assignment – 4

1 Explain DMA operation.

2 Explain How input/output can be performed using interrupts.

3 Enlist any two examples of external interrupt.

4 Content Addressable Memory.

5 Discuss associative mapping and direct mapping in organization of cache memory

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6 Explain any two interconnection structures that make it possible to form a multiprocessor system, with diagrams.

7 Discuss the differences between tightly-coupled multiprocessor and loosely-coupled multiprocessor.

8 What is virtual memory? Explain relation between address space and memory space in virtual memory system.